onero Robotic Arm A1 Product Manual

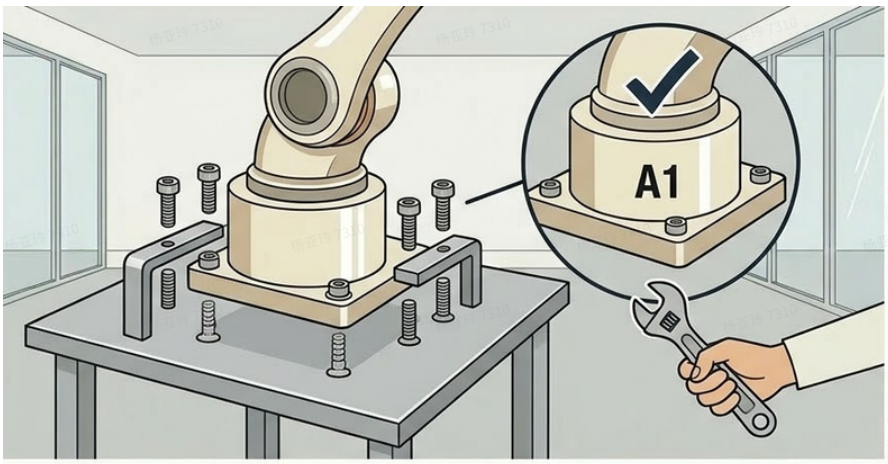
Safety Guide

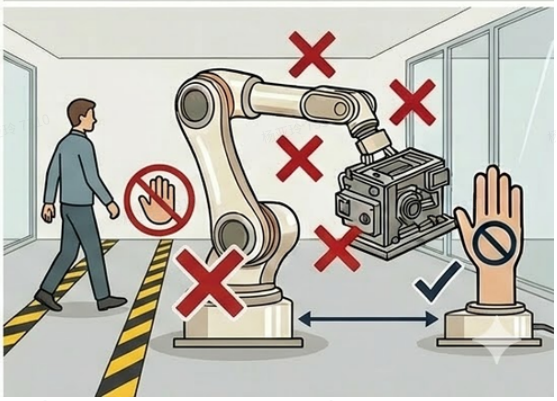
> ⚠ **WARNING:** Read this guide carefully before operating the onero Robotic Arm A1 (hereinafter referred to as the "A1") and assume all risks associated with its use.

Safe Installation

When installing the A1, secure it firmly to a flat, stable surface using screws or clamps. Do not install near fragile items, flammable materials, or in humid environments. Avoid placing the robot in high-traffic corridors or pathways.

Note: All images in this document are for illustrative purposes only and do not represent the actual appearance of the A1.

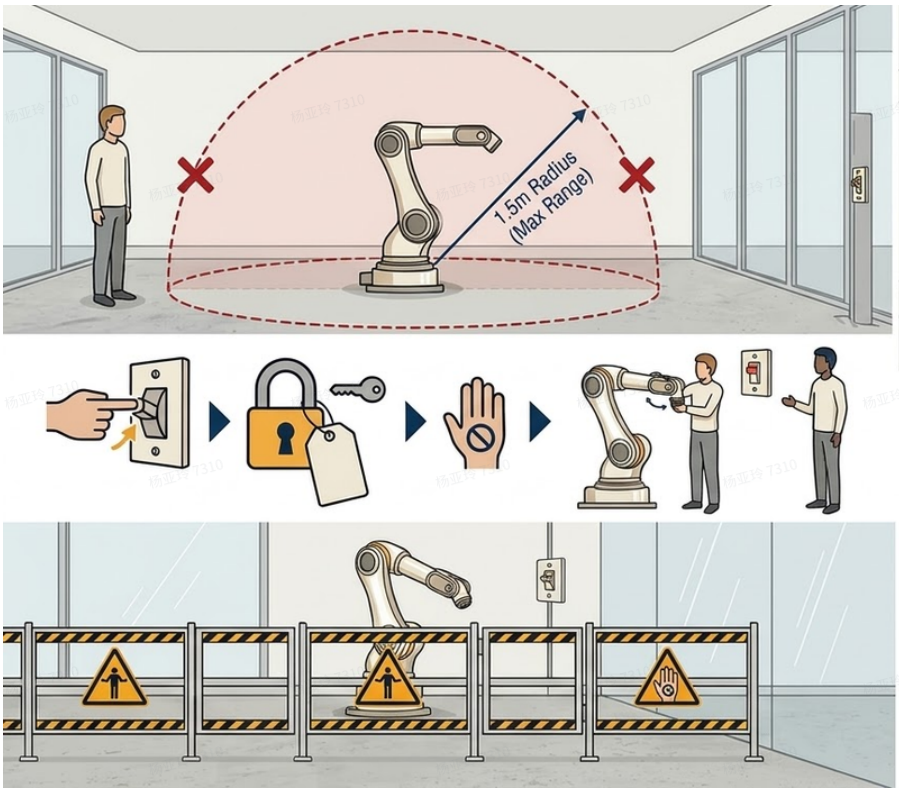


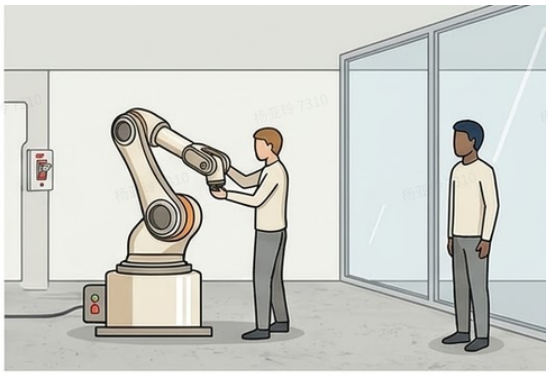


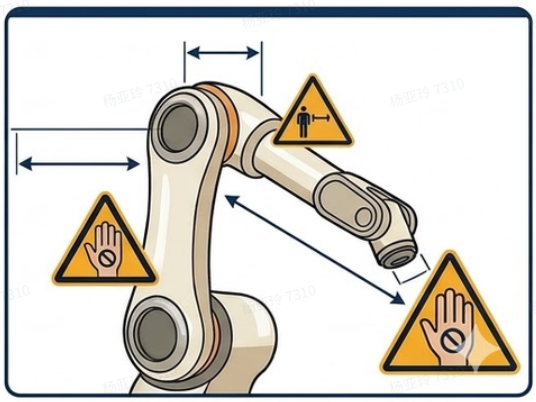
Maintaining a Safe Distance

When operating the A1, keep all body parts and objects outside the safety zone — defined as a 1.5-meter (4.9 ft) radius sphere centered on the shoulder joint. If you need to approach the robot for adjustments, first disconnect the power and ensure that nearby personnel cannot accidentally restore it. Use barriers or signage to restrict access to the operating area when necessary.

⚠ **WARNING:** When approaching the robotic arm, maintain a safe distance from protruding parts such as the elbow, shoulder, and end effector to avoid injury.

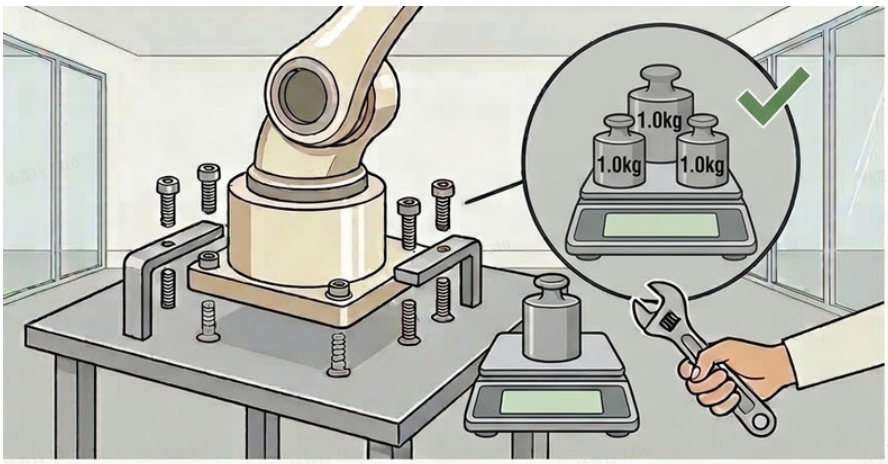


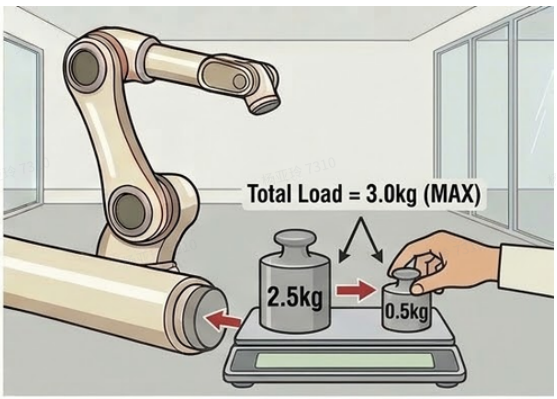


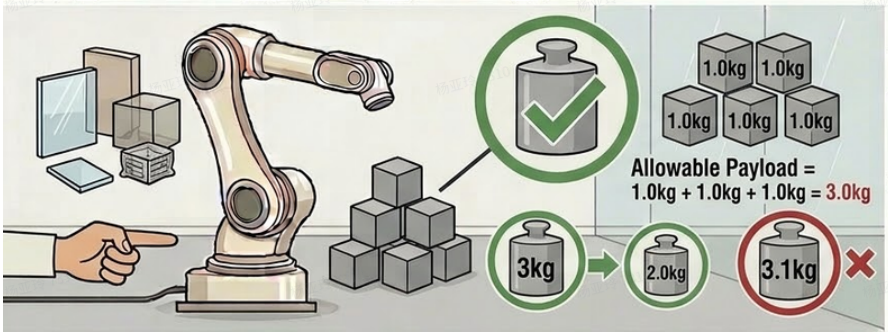


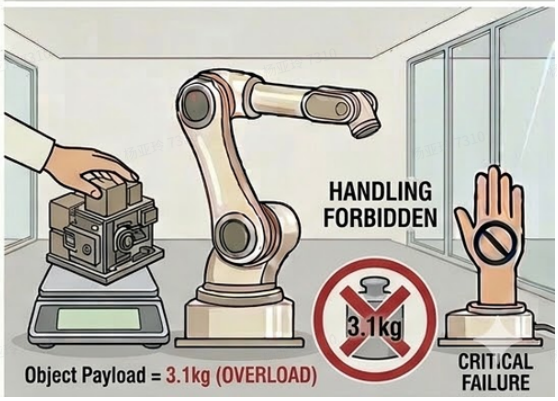
Operating Within Rated Load Limits

Do not allow the actual load on the A1 to exceed its rated maximum payload during operation. When using an end-effector, ensure the payload does not exceed the rated capacity of that end-effector.









Disclaimer

Thank you for choosing the A1 Robotic Arm. Please read and fully understand this disclaimer before using the product. By using this product, you acknowledge that you have read, understood, and accepted all of the following terms and conditions.

1. **Operating Standards and Safety Responsibility**

Users must strictly follow the accompanying user manual, installation guide, and safety instructions when operating the A1. We assume no legal liability for any losses, property damage, or personal injury caused by failure to follow instructions, improper installation, unauthorized operation, or unauthorized modification.

2. **Operating Environment and Conditions**

Users must ensure that the installation and operating environment meets all specified requirements (e.g., a level and stable mounting surface, free from flammable materials and humid conditions). Users bear full responsibility for equipment failures or safety incidents caused by unsuitable environments, unstable power supplies, or external interference.

3. **Payload and Accessory Limitations**

Users must not allow the load on the A1 to exceed its rated maximum payload of 3 kg. We bear no responsibility for any consequences resulting from overloading or from the use of incompatible third-party accessories.

4. **Modification and Repair**

Unauthorized disassembly, modification, or repair of the A1 is strictly prohibited without prior written authorization from us. We assume no liability for equipment damage, performance degradation, or safety incidents resulting from unauthorized alterations or repairs.

5. **Limitation of Liability**

To the maximum extent permitted by applicable law, we shall not be liable for any direct, indirect, incidental, special, or punitive damages arising from the use of or inability to use this product, including but not limited to loss of profits, business interruption, data loss, or other economic losses.

6. **User Discretion**

The specific application scenarios and operating methods of the A1 are determined solely by the user. Users bear full responsibility for all activities conducted using the A1 and their consequences.

7. **Risk Notice**

Robotic products may present unforeseeable risks during operation. Users are responsible for implementing appropriate safety measures (e.g., installing safety barriers and posting warning signs). Users bear full responsibility for accidents resulting from failure to take necessary precautions.

If you have any questions regarding this disclaimer or require technical support, please contact our customer service team.

I. Product Overview

The A1 is a high-precision, 7-DOF (seven-degree-of-freedom) lightweight collaborative robotic arm designed for desktop and workstation fixed-mount applications. It is well suited for industrial assembly, laboratory operations, AI training, and warehouse sorting tasks. With a compact form factor and rapid response, the A1 supports hybrid position–force control for safe and flexible human–robot collaboration.

II. Key Technical Specifications

Parameter	Description
Product Name	onero Robotic Arm A1
Degrees of Freedom	7-axis single arm
Arm Weight	4.3 kg
Maximum Payload	3 kg
Maximum Reach	739.45 mm
Repeatability	±0.1 mm
Power Supply	DC 48 V
Communication Interface	CAN 2.0, 1 Mbps
Rated Power Consumption	< 300 W
Control Modes	Position control + force control

III. Joint Parameters

Joint	Range of Motion	Maximum Speed	Rated Torque	Maximum Torque
J1	-60° ~ +180°	180°/s	15 N·m	30 N·m
J2	-180° ~ +15°	180°/s	15 N·m	30 N·m
J3	±158°	180°/s	15 N·m	30 N·m
J4	±110°	180°/s	15 N·m	30 N·m
J5	±158°	180°/s	3 N·m	7 N·m
J6	±90°	180°/s	3 N·m	7 N·m
J7	±158°	180°/s	3 N·m	7 N·m

Note: Joints J1–J4 are upper-arm joints that provide primary range of motion and load capacity. Joints J5–J7 are wrist joints that enable flexible orientation adjustment of the end-effector.

IV. Typical Application Scenarios

- **Industrial Assembly:** Precision insertion, alignment, and fastening operations.
- **Scientific Research:** Grasping, sample transport, and other repetitive precision tasks in AI-for-Science workflows.
- **Algorithm Training Platform:** Serves as the core actuator in standardized task stations, supporting high-frequency teleoperation and data collection.
- **Warehouse Sorting:** Product grasping, classification, and palletizing at fixed stations, with high positional repeatability.

V. Software and Development Support

- **Operating System:** Ubuntu 22.04 LTS.
- **Development APIs:** Open-source Python and C++ SDKs for rapid integration and custom development.
- **ROS Support:** Official ROS 2 Humble driver package, compatible with the MoveIt 2 motion planning framework.
- **Simulation Environment:** Gazebo simulation support with URDF models and controller configuration files.

VI. Maintenance and Care

- **Cleaning:** Wipe the housing with a dry soft cloth or a slightly damp lint-free cloth. Do not use corrosive solvents such as alcohol or acetone.

Status Indicators:

- LED solid green: System enabled — normal standby or operating.
- LED solid red: System disabled or fault detected (e.g., over-temperature, over-current). Check logs or restart.

VII. Packing List

This robotic arm product is named: onero Robotic Arm A1. The suffixes L, R, and S correspond to left arm, right arm, and dual arm respectively.

Product	onero Robotic Arm A1L	onero Robotic Arm A1R	onero Robotic Arm A1S
Items Included	A1 Left Arm Unit × 1	A1 Right Arm Unit × 1	A1 Dual Arm Unit × 1
	USB-to-CAN adapter (for CAN debugging) × 1	USB-to-CAN adapter (for CAN debugging) × 1	USB-to-CAN adapter (for CAN debugging) × 2
	Micro USB Cable × 1	Micro USB Cable × 1	Micro USB Cable × 2
	Power & CAN Communication Cable × 1	Power & CAN Communication Cable × 1	Power & CAN Communication Cable × 2
	Mounting Screws × 5	Mounting Screws × 5	Mounting Screws × 10
	Hex Key × 1	Hex Key × 1	Hex Key × 1

VIII. Operating Environment Requirements

- **Operating Temperature:** 5°C to 50°C (41°F to 122°F)
- **Relative Humidity:** 35% RH to 80% RH (non-condensing).
- **Mounting Method:** Desktop or base mounting. The mounting surface must be level, stable, and free from vibration.

IX. Precautions

- Ensure the power supply meets the minimum requirement of DC 48 V / 6 A to prevent performance degradation or unexpected shutdown.
- Before the initial power-on, confirm that the robotic arm's full range of motion is clear of obstacles.
- Avoid executing high-speed force-control operations without an end-effector force protection strategy in place, as doing so may damage the gripper or workpiece.
- Do not disassemble the joint housings, as this may compromise the sealing integrity and calibration accuracy of the joints.

X. Support and Warranty

- **Warranty:** 6-month limited warranty from delivery date (excluding user-caused damage).

Technical Support:

- Email: support@switch-bot.com
- China: 400-822-5888 (UTC+8 10:00-12:20, 14:00-22:00, except public holidays)
- North America: +1 866-530-2887 (UTC-4 10:00-19:00, except public holidays)
- Japan: 0120-676-162 (UTC+9 11:00-13:20, 14:40-23:00, except public holidays)
- Other regions: Contact us via email or the SwitchBot official app.

Documentation and Community Resources:

- Documentation Center: https://github.com/OpenWonderLabs/RobotDoc
- SDK (C/C++ and Python): https://github.com/OpenWonderLabs/ArmApi
- ROS 2 Driver: https://github.com/OpenWonderLabs/ArmRos